

Nathan Espejo

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EDUCATION

Western University

Bachelor of Engineering Science in Software Engineering

London, ON

Aug. 2024 – May 2028

EXPERIENCE

Software Developer

Sept 2025 – Present

3D Western

London, ON

- Developing a Dockerized 3D printing pipeline for a 24-printer farm, automating model slicing, orientation, support generation, and printability validation for a free campus printing service.
- Built data generation workflows for 3D model analysis by scraping assets and creating automated rendering scripts to produce labeled image datasets for AI training.

Software Engineering Intern

May 2025 – Aug 2025

TeraGo

Markham, ON

- Built an automated Jira ticket workflow from email notifications using a local LLM (Ollama) for structured extraction, Python orchestration, and Microsoft Graph for secure message retrieval.

IT Intern

May 2025 – Aug 2025

Microcomputer Consulting Inc.

Mississauga, ON

- Configured and deployed client workstations across multiple organizations, including Windows imaging, account setup, and environment-specific software configuration.

PROJECTS

Local AI Platform | *Docker, Linux, Tailscale, Bifrost, MCP, llama.cpp*

Apr 2026

- Architected a fully dockerized local-first AI stack with secure remote access, hybrid model routing, and custom MCP servers for tool-enabled agent workflows.
- Built course-specific retrieval pipelines by parsing large PDFs, extracting relevant chapters, chunking and embedding content, and grounding agent responses for study workflows.

AgenticArmy | *TypeScript, Python, FastAPI, VS Code Extension*

Jan 2026

- Built a human-in-the-loop multi-agent coding workflow with a VS Code extension and backend coordinator for planning, conflict analysis, and QA.
- Implemented orchestration patterns including approval gates, state transitions, event-driven coordination, and simulation mode for reliable agent execution.

Auralis | *Next.js, React, Three.js, FastAPI, face-api.js*

Nov 2025

- Built an AI-powered virtual doctor with emotion-aware consultations and report generation, winning **Best AI Application Built with Cloudflare**.
- Developed emotion mismatch detection and adaptive response logic using facial signals and conversational context to improve safety-focused interactions.

EyeCandy | *TypeScript, FastAPI, Shopify, AR, Snap Spectacles*

Sept 2025

- Developed an AR shopping workflow that visualizes products on users in real time, winning the **Snap: Spectacles AR Runner Up Prize** at Hack the North 2025.
- Built product data and 3D asset pipelines that convert storefront inputs into AR-ready experiences for rapid prototyping in a hackathon setting.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C#, C++, Java, Kotlin, SQL, Bash

AI Systems & ML: Local/Hybrid LLM Workflows, Agentic Orchestration, MCP, RAG, Computer Vision, PyTorch, TensorFlow, scikit-learn, pandas

Full-Stack Development: React, Next.js, FastAPI, Node.js, REST APIs, MongoDB, Supabase, MySQL

Infrastructure & Tools: Linux (Arch, Debian), Docker, Git, Tailscale, Cursor, GitHub Copilot, Claude Code, VS Code, Unity3D, Blender, OnShape

Embedded & Hardware: Arduino, ESP32, Raspberry Pi, PlatformIO, BLE, 3D Printing, CAD Prototyping